In-plane dielectric constant and conductivity of confined water

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Water is essential for almost every aspect of life on our planet and, unsurprisingly, its properties have been studied in great detail[1]. However, disproportionately little remains known about the electrical properties of interfacial and strongly confined water[2,3] where its structure deviates from that of bulk water, becoming distinctly layered[4,5]. The structural change is expected to affect water's conductivity and particularly its polarizability, which in turn modifies intermolecular forces that play a crucial role in many physical and chemical processes[6-9]. Here we use scanning dielectric microscopy[10] to probe the in-plane electrical properties of water confined between atomically flat surfaces separated by distances down to 1 nm. For confinement exceeding a few nm, water exhibits an in-plane dielectric constant close to that of bulk water and its proton conductivity is notably enhanced, gradually increasing with decreasing water thickness. This trend abruptly changes when the confined water becomes only a few molecules thick. Its in-plane dielectric constant reaches giant, ferroelectric-like values of about 1,000 whereas the conductivity peaks at a few S·m^{-1}, close to values characteristic of superionic liquids. We attribute the enhancement to strongly disordered hydrogen bonding induced by the few-layer confinement, which facilitates both easier in-plane polarization of molecular dipoles and faster proton exchange. This insight into the electrical properties of nanoconfined water is important for understanding many phenomena that occur at aqueous interfaces and in nanoscale pores.

In the bulk and at room temperature (RT), water exhibits exceptionally high dielectric constant ($\varepsilon_{bulk} \approx 80$) and high (for a wide-bandgap insulator) electrical conductivity ($\sigma_{bulk} \approx 10^{-5}$ S·m^{-1})[1]. Both characteristics are inherently connected with the ability of water molecules to form hydrogen bonds[11-13] and are key to water's main properties. Among them is its remarkable ability to dissolve more substances than any other liquid[7,8], which stems from water's high ε_{bulk} that efficiently suppresses Coulomb interactions between solutes. The strong dielectric screening is also critical in biochemical processes responsible for life[7] (e.g., proteins' folding and assembly, their interaction with nucleic acids, and ion transport across cell membranes). Importantly, a multitude of phenomena involving water (including solvation) occurs at solid interfaces where liquid water exhibits a distinct layered structure[4,5]. Within these few layers, the hydrogen-bond network is greatly altered as compared to that of bulk water, no longer following the ice rules[14]. Accordingly, the electrical properties of water near surfaces and inside nanoscale cavities are expected to be different from those of bulk water[2,3]. These differences have been the subject of intense research. In particular, proton conductivity of near-surface water has been reported to become much larger than σ_{bulk}, although the magnitude of this enhancement and the underlying mechanism remain debated[15-19]. Meanwhile, it has proven difficult to assess the dielectric properties of interfacial and nanoconfined water[20,21]. Only recently[22] it has been shown that, in the direction perpendicular to surfaces, water within a few near-surface layers becomes nonpolarizable, exhibiting the dielectric constant $\varepsilon_{\perp} \approx 2$, in agreement with many theoretical predictions[23-26]. However, the in-plane dielectric constant $\varepsilon_{//}$ of interfacial water remains essentially unknown, because $\varepsilon_{//}$ is even more challenging to measure than $\varepsilon_{\perp}$ for the lack of suitable experimental techniques. The in-plane polarizability of near-surface water is also poorly understood theoretically and, depending on assumptions and confinement strength, $\varepsilon_{//}$ was predicted to attain values from giant to bulk-like to exceptionally small[27-31].

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Scanning dielectric microscopy on water-filled nanochannels

To probe the in-plane response of nanoconfined water, we have employed scanning dielectric microscopy (SDM)[10]. Our setup is schematically shown in Fig. 1a and detailed in Methods. Briefly, nanoscale channels of various heights h and having the width of ~ 200 nm were fabricated by van der Waals assembly of atomically flat monocrystals of hexagonal boron nitride (hBN) and graphite (for fabrication details, see Methods). An AC voltage was applied to an AFM tip, and its mechanical response to the electrostatic force translated into local electrical impedance. By scanning the tip across the channels, the dielectric response of water inside the nanochannels was measured and compared with that of nearby insulating spacers made of hBN with the known dielectric constant ($\varepsilon_{hBN} \approx 4$). The approach is conceptually similar to that used to study $\varepsilon_{\perp}$ of nanoconfined water[22] but the previous setup was insensitive to $\varepsilon_{//}$ because the electric field E was applied perpendicular to the channels. To measure $\varepsilon_{//}$, we have introduced two main changes. First, the ground electrode (graphite crystal in Fig. 1a; Extended Data Fig. 1) has been moved away from the probed water layer by adding a relatively thick hBN crystal to the assembly, which creates an in-plane field component $E_{//}$ inside the water (Fig. 1a). Second, we have expanded the measurement bandwidth by ~ 5 orders of magnitude from the standard kHz frequencies into the GHz regime, which is challenging but essential for measuring $\varepsilon_{//}$ in the presence of unexpectedly high $\sigma_{//}$ found for few-layer (quasi-2D) water (see below).

Representative SDM images from our relatively large (reference) channels with $h \approx 30$ nm (Fig. 1b) are shown in Figs. 1c and 1d before and after filling them with deionized water, respectively (also see Extended Data Fig. 2). The dielectric contrast between empty channels (ε = 1) and hBN spacers appears negative as expected, reflecting the relatively larger dielectric constant of hBN. After the channels were filled, the contrast changed and became strongly positive, confirming the presence of water inside the channels and its strong dielectric response. With increasing the measurement frequency, f, the contrast for empty channels did not change, again as expected (Fig. 1c). For water-filled channels, it remained positive with increasing f from kHz to MHz and GHz but weakened (Fig. 1d). For each f, we then plotted the average contrast of water with respect to hBN (Fig. 1e) and, by repeating the imaging at many different f (see detailed measurement and calibration

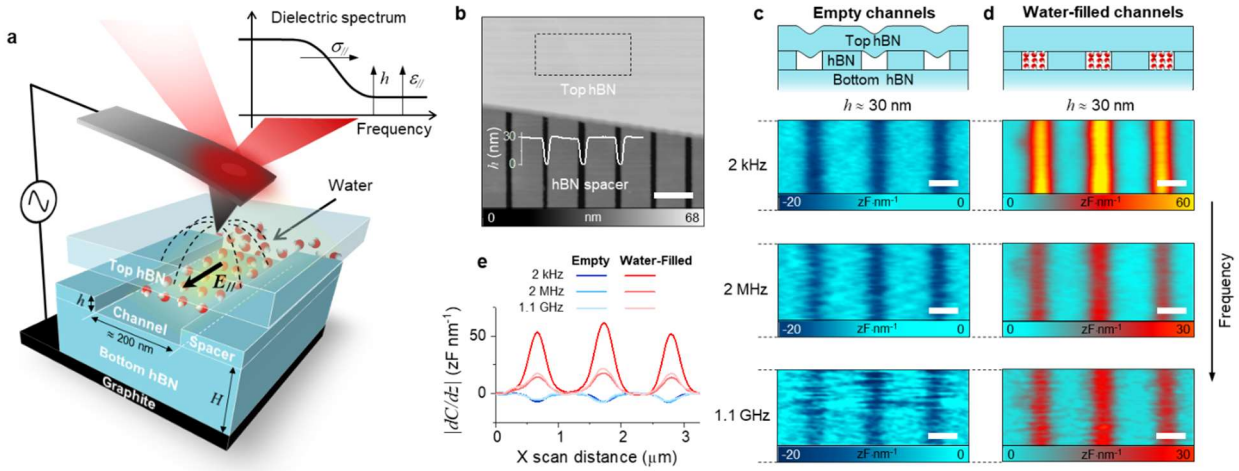

Fig. 1| Broadband dielectric imaging of nanoconfined water. (**a**) Schematics of the experimental setup and our devices. The hBN top, bottom and spacer layers are shown in light blue and the graphite ground electrode in black. The water is filled from a backside inlet (Extended Data Fig. 1d). Oscillating voltage (f from 0.1 kHz to 1.1 GHz) is applied to the AFM tip, and the measured local impedance is converted into dielectric spectra such as that schematically shown in the inset. Relatively thick bottom hBN ($H \approx 50$–200 nm) is used to create in-plane field $E_{//}$ inside water channels. (**b**) AFM micrograph of a device with $h \approx 30$ nm with the inset showing a trace over the exposed spacers. (**c** and **d**) Dielectric images at characteristic f of 2 kHz, 2 MHz and 1.1 GHz for the same three channels before and after filling them with water, respectively. The images were taken at 295 ± 1 K from the area outlined in (b) by the dashed rectangle. The top panels in (c) and (d) sketch the channels' cross-section (not to scale). The slight sagging of the top hBN into empty channels disappears after filling them with water. The sagging was monitored by AFM to ensure that the channels were fully filled[22]. (**e**) Dielectric profiles (averaged over 10 lines) from (c,d); color coded. $|dC/dz|$ is defined in Methods. Scale bars: (b) 1 μm, (c-d) 500 nm.

procedures in Methods; Extended Data Fig. 3), we obtained dielectric dispersion curves such as those shown in Fig. 2a (see the blue symbols for $h \approx 30$ nm).

Although the experimental curve resembles the Debye dispersion, the orientational relaxation of water dipoles responsible for the Debye behavior of bulk water plays little role at $f < 10$ GHz, and the dispersion found at our relatively low f is caused by conductive losses (Methods). Indeed, at high f, the dielectric response is determined by water's $\varepsilon_{//}$ and is independent of $\sigma_{//}$, which yields the high-f plateau with its strength increasing with increasing $\varepsilon_{//}$ and h. Water's $\sigma_{//}$ starts contributing at f below the conductivity relaxation frequency $f_r = \sigma_{//}/2\pi\varepsilon_0\varepsilon_{//}$, giving rise to a stronger dielectric contrast. The conductivity contribution is cut off at $f_c = \alpha\sigma_{//}/2\pi\varepsilon_0$ by capacitive coupling between the water and the bottom graphite as well as the AFM tip, which results in the low-f plateau. The cut-off frequency f_c increases proportionally to $\sigma_{//}$ and is independent of $\varepsilon_{//}$, where α is the factor that depends on h and the measurement geometry. The inset of Fig. 1a summarizes the tendencies expected for the dielectric response with changing h, $\varepsilon_{//}$ and $\sigma_{//}$. Further details about SDM measurements are provided in Methods (Extended Data Figs. 4-10).

Dielectric dispersion of nanoconfined water

In addition to the reference spectrum from the 30-nm channels, Fig. 2a shows the spectra obtained for ultrathin water in channels with other representative heights $h \approx 5$ and 1.5 nm (cyan and red) whereas Fig. 2b provides examples of their SDM imaging (also see Extended Data Fig. 4). One can see two clear trends in how the dielectric response of water evolves with decreasing h. First, the transition between low- and high-f plateaus shifts to higher frequencies, from kHz to MHz. This is also apparent from the images of Fig. 2b where changes in the contrast occur at higher f for the thinner channel. The behavior unambiguously shows that water becomes more conductive under strong confinement. This is somewhat expected because water is known to interact with hBN walls, which results in near-surface dissociation of water molecules and higher proton conductivity[17,32]. The strongly enhanced σ is also consistent with the previous results for water inside hBN nanochannels and nanoporous media[33,34]. Second, the high-f dielectric response does not gradually diminish with decreasing h as expected if ε of nanoconfined water were to remain the same as of bulk water

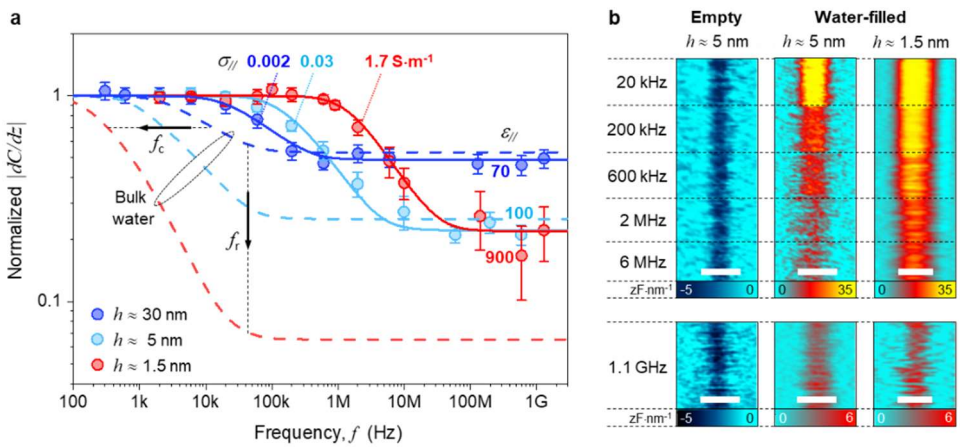

Fig. 2| Dielectric dispersion of nanoconfined water. (**a**) Examples of the dielectric spectra for three representative h. The color-coded symbols are the peak values for $|dC/dz|$ curves as in Fig. 1e, normalized by their magnitude at low f. Bars: standard error as defined in Extended Data Fig. 10. Solid curves: best fits using the numerical simulations. The fits match f_c (yielding $\sigma_{//}$) and the y-axis position of the high-f plateau (yielding $\varepsilon_{//}$). Dashed curves: same simulations for bulk water (ε_{bulk} = 80 and σ_{bulk} = $2{\cdot}10^{-4}$ S·m^{-1}; see Methods). The black lines show the evolution of f_c and f_r with decreasing h. (**b**) Dielectric images taken for individual channels with $h \approx 1.5$ and 5 nm (empty and filled with water), as specified in the legends. Upper panels: images for a fixed-y position of the AFM tip; ~ 25 lines were recorded at each f. Lower panels: images at 1.1 GHz, ~ 100 lines are shown for clarity because of higher noise. Scale bars, 500 nm.

(cf. inset of Fig. 1a). Instead, the high-f plateaus for $h \approx 1.5$ and 5 nm in Fig. 2a remain of similar height. This observation qualitatively implies that water's ε notably increases under strong confinement. The positive contrast at high f for $h \approx 1.5$ nm is particularly striking if compared with the behavior found by SDM without a parallel component of E. In the latter case[22], the dielectric contrast reversed from positive to negative at $h \approx 3$ nm because of diminishing $\varepsilon_{\perp}$. This does not happen for the in-plane measurement geometry even for our smallest $h \approx 1$ nm, and again implies large $\varepsilon_{//}$ for water inside atomic-scale channels.

The measured spectra are essentially determined by the two characteristic frequencies f_r and f_c that in turn reflect $\varepsilon_{//}$ and $\sigma_{//}$ of nanoconfined water, respectively. Accordingly, changes in the frequencies with varying h can be translated into relative changes of the corresponding electrical characteristics. With reference to the inset of Fig. 1a, the experimental spectra in Fig. 2a allow one to conclude immediately that both $\varepsilon_{//}$ and $\sigma_{//}$ increase with decreasing the water thickness. However, to find the scale of these changes and the absolute values of $\varepsilon_{//}$ and $\sigma_{//}$ requires knowledge of the factor α that depends on h and other geometric parameters. To this end, we used full-3D numerical simulations and calculated the detailed electric field distribution across our nanochannels. From this analysis, we obtained anticipated dielectric dispersions for different h (Extended Data Figs. 5 and 6). Then, we determined $\sigma_{//}$ and $\varepsilon_{//}$ by fitting f_c and positions of the high-f plateau (all other parameters were found experimentally; Extended Data Fig. 10). These simulations also took into account that the dielectric response of nanoconfined water is anisotropic, using the $\varepsilon_{\perp}(h)$ dependence reported previously[22]. Examples of this analysis are presented in Fig. 2a by the solid curves. For comparison, the dashed curves in the same figure represent the behavior expected if the confined water retained the electrical properties of bulk water. In addition, we analyzed the experimental spectra using two analytical approximations where i) the AFM tip was simulated as a point charge and the nanochannel as a laterally infinite multilayer stack (Extended Data Fig. 8) and ii) the measurement geometry was simulated using an equivalent electrical circuit (Extended Data Fig. 9). This provided an approximation for α as a function of the main parameters, $\alpha \cong (h \cdot w/l^*) \cdot [1/(2\pi R) + H/(\varepsilon_{hBN} \cdot w \cdot l^*)]$, where R is the AFM tip radius and l^* is the effective channel length (Methods). Although less accurate, the latter models were found to yield close values for both $\varepsilon_{//}$ and $\sigma_{//}$. Furthermore, we verified that the extracted values depended little on $\varepsilon_{\perp}$ (because $\varepsilon_{//}$ is much larger, see Methods, Extended Data Fig. 6f and Extended Data Fig. 7) and were independent of $\sigma_{\perp}$ (the employed measurement geometry is insensitive to the latter, see Extended Data Fig. 6e). The use of several approaches have shown that our results are robust with respect to modeling and, also, corroborated our qualitative conclusions drawn from the spectral shifts, as discussed above.

We have studied many nanochannel devices with h ranging from ~1 to 60 nm and, using f from a few hundred Hz to GHz, obtained their dielectric spectra similar to those shown in Fig. 2a. Then applying the full numerical analysis for each device, water's $\varepsilon_{//}$ and $\sigma_{//}$ were extracted as a function of h. The results are summarized in Fig. 3.

In-plane dielectric constant

Inside our channels with $h \geq 5$ nm, water was found to exhibit $\varepsilon_{//} \approx 74 \pm 17$, that is, little difference with respect to bulk water (Fig. 3a). The dielectric constant suddenly changed for quasi-2D water ($h \approx 1$-2 nm) that showed a sharp increase in $\varepsilon_{//}$ by an order of magnitude, reaching 1,030 ± 350. Such large dielectric constants are typical of ferroelectrics. This behavior is in stark contrast to that of $\varepsilon_{\perp}$ for the same range of thicknesses[22]. The latter constant exhibits a decrease by a factor of ~40 so that quasi-2D water becomes essentially nonpolarizable in the perpendicular direction[22]. Although the giant $\varepsilon_{//}$ for water under atomic-scale confinement may seem surprising, the result agrees (at least qualitatively) with most simulations that have predicted enhanced $\varepsilon_{//}$ and strong dielectric anisotropy for water in the vicinity of solid interfaces[23,24,27-30]. The phenomenon can be understood by noting that, while the interaction with solid surfaces hinders water dipoles' ability to align along E applied in the perpendicular direction, the dipoles are still relatively free to rotate and align in the in-plane direction.

For large channels, it is reasonable to consider confined water as consisting of an inner layer of bulk water molecules unaffected by the confinement and two near-surface layers with a different interfacial dielectric constant $\varepsilon_{//\text{int}}$, as sketched in the inset of Fig. 3a. This simple model (three capacitors in series) has previously proven successful in describing the observed $\varepsilon_{\perp}(h)$ dependence qualitatively well and suggested the presence of an 'electrically dead' near-surface layer[22]. Its thickness h_{int} was estimated as ~7 Å, in agreement with the thickness of layered

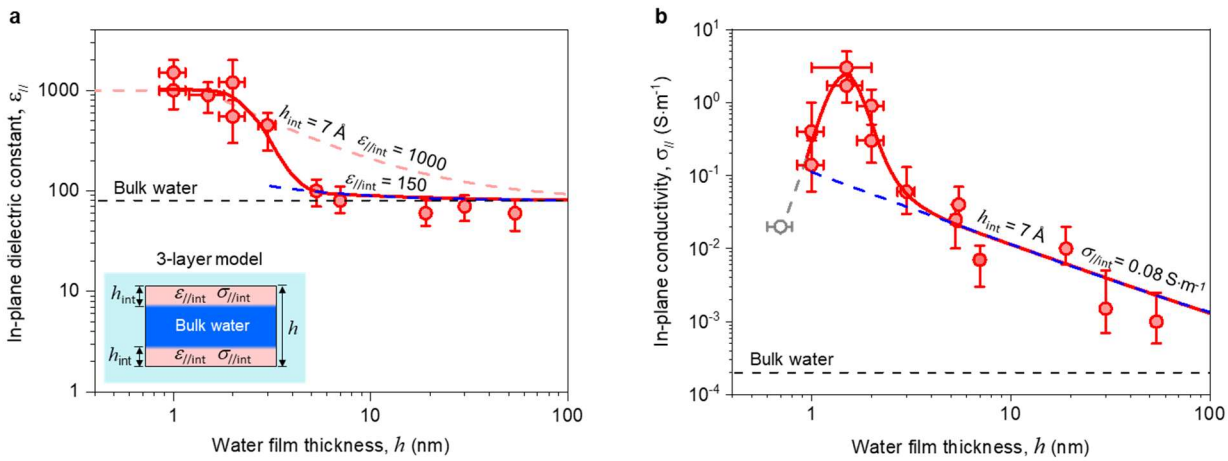

Fig. 3| In-plane electrical properties of water under molecular-scale confinement. (**a**) Experimental data for $\varepsilon_{//}$ are shown by red symbols. Solid curve: guide to the eye. Long-dash curves: 3-layer capacitance model explained in the inset. Shown in light red and blue are the model's fits with $\varepsilon_{//\text{int}}$ = 1,000 and 150, respectively, using the interfacial-layer thickness h_{int} found in Ref.[22] Black horizontal line, $\varepsilon_{\text{bulk}}$. (**b**) Red symbols and red curve: same as in (a) but for σ. Grey symbol: experimental value incorporated from Ref.[33] Black line: bulk conductivity of the used water. Blue line: best fit for $h \geq$ 4 nm using the 3-layer model in (a) and assuming different conductivities for bulk and near-surface water. The error bars in both panels are the uncertainty as defined in Extended Data Fig. 10.